



EP 201: Reverse Engineering

Course Code	EP 201
Course Name	Reverse Engineering
L-T-P-C	0-0-2-1
Prerequisites	None (Discipline Core)
Intended for	B.Tech Engineering Physics
Mutual Exclusion	None
Approval	60th BoA (Jun 2025)

Preamble

Reverse engineering is a valuable skill required for engineers, experimentalists, software developers and anyone who wants to know how things work. In this course, the focus will be on the reverse engineering of either hardware products or debugging of source code or both. After the completion of the course, students should be able to

- Understand the methodology involved in reverse engineering
- Dismantle and assemble a product
- Test, analyze and evaluate a product
- Propose a better version to create the product.
- Document the process involved and reporting.

Course Modules

During the course, the students will be given the task of

(a) Dismantling and assembling of hardware like

- electrical appliances (that is working), motors, vacuum pumps etc. to study its internal make. Similar appliances will also be given that are not working and they have to identify the fault in it and make it working.
- mechanical system like a bicycle.
- instrumental set up that is used to study electronic, optical, magnetic properties etc.
- measuring devices and source meters like multimeter, voltage source, current source etc.

(b) Studying a source code and debugging it, fixing the errors.

References

1. Wego Wang, *Reverse Engineering: Technology of Reinvention*.